where, R is capacity ratio of deadweight of the ship (tonnes) as determined by paragraph 2.4 divided by the total cubic capacity of the cargo tanks of the ship (m3).

.13 Length between perpendiculars , Lpp means 96 per cent of the total length on a waterline at 85 per cent of the least moulded depth measured from the top of the keel, or the length from the foreside of the stem to the axis of the rudder stock on that waterline, if that were greater. In ships designed with a rake of keel the waterline on which this length is measured should be parallel to the designed waterline. The length between perpendiculars (L pp) should be measured in metres.

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parallel to the designed waterline. The length between perpendiculars (L pp) should be measured in metres.

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APPENDIX 1

A GENERIC AND SIMPLIFIED MARINE POWER PLANT

Note 1: Mechanical recovered waste energy directly coupled to shafts need not be measured, since the effect of the technology is directly reflected in the Vref .

Note 2: In case of combined PTI/PTO, the normal operational mode at sea will determine which of these to be used in the calculation.

AUXILIARY
ENGINES

BOILER

CARGO HEAT

THRUSTERS

CARGO PUMPS

REEFERS

CARGO GEAR

BALLAST PUMPS
SWITCH BOARD
SHAFT MOTOR PPTI

WASTE HEAT
RECOVERY etc.
PAEeff
MAIN ENGINE PME SHAFT POWER
PS
MAIN ENGINE
PUMPS (2.5% PME)
ACCOMMODATION
(250 kW)
PAE
SHAFT
GENERATOR PPTO
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ACCOMMODATION
(250 kW)
PAE
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GENERATOR PPTO
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APPENDIX 2

GUIDELINES FOR THE DEVELOPMENT OF ELECTRIC POWER TABLES
FOR EEDI (EPT-EEDI)

1 Introduction to the document "Electric Power Table for EEDI"

1.1 This appendix contains a guideline for the document "Electric Power Table for EEDI" which is similar to the actual shipyards' l oad balance document, utilizing well defined criteria, providing standard format, clear loads definition and grouping, standard load factors, etc. A number of new definitions (in particular the "groups") are introduced, giving an apparent greater complexity to the calculation process. However, this intermediate step to the final calculation of PAE stimulates all the parties to a deep investigation through the global figure of the auxiliary load, allowing comparisons between different ships and technologies and eventually identifying potential efficiencies improvements.

2 Auxiliary load power definition

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SOURCE: MEPC.212(63).pdf
the auxiliary load, allowing comparisons between different ships and technologies and eventually identifying potential efficiencies improvements.

2 Auxiliary load power definition

2.2 PAE is to be calculated as indicated in paragraph 2.5.6 of the Guidelines, together with the following additional three conditions:

.1 no emergency situations (e.g. "no fire", "no flood", "no blackout", "no partial blackout");

.2 evaluation time frame of 24 hours (to account loads with intermittent use); and

.3 ship fully loaded of passenger and/or cargo and crew.

3 Definition of the data to be included in the Electric Power Table for EEDI

3.1 The Electric power table for EEDI calculation should contain the following data elements, as appropriate:

.1 Load's group;
.2 Load's description;
.3 Load's identification tag;
.4 Load's electric circuit Identification;
.5 Load's mechanical rated power " Pm" [kW];
.6 Load's electric motor rated output power [kW];
.7 Load's electric motor efficiency " e" [/];
.8 Load's Rated electric power " Pr" [kW];
.9 Service factor of load " kl" [/];
.10 Service factor of duty " kd" [/];
.11 Service factor of time " kt" [/];

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SOURCE: MEPC.212(63).pdf
.7 Load's electric motor efficiency " e" [/];
.8 Load's Rated electric power " Pr" [kW];
.9 Service factor of load " kl" [/];
.10 Service factor of duty " kd" [/];

.11 Service factor of time " kt" [/];
.12 Service total factor of use " ku" [/], where ku=kl·kd·kt;
.13 Load's necessary power "Pload" [kW], where Pload=Pr·ku;
.14 Notes;
.15 Group's necessary power [kW]; and
.16 Auxiliaries load's power P AE [kW].

4 Data to be included in the Electric Power Table for EEDI

Load groups

4.1 The Loads are put into defined groups, allowing a proper breakdown of the auxiliaries. This eases the verification process and makes it possible to identify those areas where load reductions might be possible. The groups are listed below: